Data Model Exercise Solutions

Chapter: Data warehouse modeling (Kimball) is based off of 2 types of tables: Fact and dimensions

1. What are the fact tables in our TPCH data model?
2. `lineitem`
    1. Contains transactional data about items within orders
    2. **Measures**: quantities, prices, discounts, taxes, shipping dates
    3. **Foreign keys to**: orders, parts, suppliers
3. `orders`
    1. Contains order-level transactional data
    2. **Measures**: total price, order priority, order status
    3. **Foreign keys to**: customers

Why they're fact tables:

1. Contain quantitative measures (prices, quantities, dates)
2. Record business transactions (sales events)
3. Have foreign keys linking to dimension tables
4. Large volume of granular transactional records
5. Support analytical queries for business metrics

The **lineitem** table is the most detailed fact table (order line level), while orders provides order-level facts. Both capture different granularities of the same business process - sales transactions.

2. What are the dimension tables in our TPCH data model?
3. `customer` - Customer information
    1. **Attributes:** name, address, phone, account balance, market segment
    2. **Purpose:** Describes who is buying
4. `supplier` - Supplier information
    1. **Attributes:** name, address, phone, account balance
    2. **Purpose:** Describes who is selling/supplying parts
5. `part` - Product/part information
    1. **Attributes:** name, manufacturer, brand, type, size, container, retail price
    2. **Purpose:** Describes what is being sold
6. `nation` - Country information

    1. **Attributes:** nation name, region key
    2. **Purpose:** Geographic dimension for customers and suppliers

7. `region` - Regional information

    1. **Attributes:** region name, comment
    2. **Purpose:** Higher-level geographic grouping of nations

Why they're dimension tables: 1. Contain descriptive attributes (names, addresses, characteristics) 2. Relatively static data (changes infrequently) 3. Provide context for fact table measures 4. Support filtering and grouping in analytical queries 5. Smaller volume compared to fact tables 6. Referenced by foreign keys from fact tables

These dimensions enable analysis by customer segments, supplier performance, product categories, and geographic regions.

3. What source tables in the TPCH data model would you consider to create a customer dimension table?

4. `customer` - Main customer data

    1. **Direct attributes:** c_custkey, c_name, c_address, c_phone, c_acctbal, c_mktsegment
    2. **Foreign key:** c_nationkey (links to nation table)

5. `nation` - Customer geographic context

    1. **Attributes:** n_name, n_regionkey
    2. **Purpose:** Provides customer's country information
    3. **Join:** customer.c_nationkey = nation.n_nationkey

6. `region` - Higher-level geographic grouping

    1. **Attributes:** r_name
    2. **Purpose:** Provides customer's regional classification
    3. **Join:** nation.n_regionkey = region.r_regionkey

Sample query to create `dim_customer`

```
%%sql
SELECT
    c.c_custkey as customer_key,
    c.c_name as customer_name,
    c.c_address as customer_address,
    c.c_phone as customer_phone,
    c.c_acctbal as account_balance,
```

```
    c.c_mktsegment as market_segment,
    n.n_name as nation_name,
    r.r_name as region_name
FROM customer c
JOIN nation n ON c.c_nationkey = n.n_nationkey
JOIN region r ON n.n_regionkey = r.r_regionkey
```